

WIESENBURG, Germany

WMO: 103680

Lat: 52.1206N

Lon: 12.4586E

Elev: 188

StdP: 99.09

Time Zone: 1.00 (EUC)

Period: 94-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.0	(c) -9.6	(d) -14.8	(e) 1.1	(f) -11.2	(g) -11.9	(h) 1.4	(i) -8.6	(j) 12.4	(k) 3.7	(l) 11.4	(m) 1.7	(n) 5.6	(o) 90	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.7	(c) 29.8	(d) 19.0	(e) 27.5	(f) 18.1	(g) 25.6	(h) 17.6	(i) 20.2	(j) 26.6	(k) 19.2	(l) 25.0	(m) 18.3	(n) 23.7	(o) 3.8	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.3	22.0	17.2	12.6	21.2	16.3	11.9	20.3	58.6	26.5	55.2	25.0	52.4	23.8	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.4	(b) 9.1	(c) 8.0	(e) -14.3	(f) 33.0	(g) 3.0	(h) 1.9	(i) -16.4	(j) 34.4	(k) -18.2	(l) 35.5	(m) -19.9	(n) 36.6	(o) -22.0	(p) 38.0		
(a) 10.4	(b) 9.1	(c) 8.0	(e) -14.6	(f) 22.0	(g) 3.0	(h) 0.8	(i) -16.7	(j) 22.6	(k) -18.5	(l) 23.1	(m) -20.1	(n) 23.5	(o) -22.3	(p) 24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-0.5	1.2	3.7	8.0	12.9	15.5	17.9	18.5	13.5	8.9	3.5	-0.3
	DBStd	7.81	4.84	4.87	3.63	4.42	3.70	3.63	3.50	3.38	3.25	3.84	3.61	4.59
	HDD10.0	1458	324	248	197	89	14	1	0	0	5	67	195	318
	HDD18.3	3654	583	481	453	310	174	101	51	41	148	293	444	576
	CDD10.0	954	0	1	2	30	102	166	246	264	111	32	1	0
	CDD18.3	109	0	0	0	1	4	16	38	46	4	0	0	0
	CDH23.3	997	0	0	0	8	51	167	355	385	31	0	0	0
	CDH26.7	274	0	0	0	1	4	50	114	103	2	0	0	0
Wind	WSAvg	3.8	4.0	4.6	4.4	3.9	3.5	3.4	3.4	3.4	3.6	3.9	3.8	4.0
Precipitation	PrecAvg	610	49	37	41	31	58	61	72	66	53	42	48	51
	PrecMax	843	96	70	98	85	127	126	203	150	146	109	103	102
	PrecMin	427	5	9	7	1	7	26	17	20	14	8	1	15
	PrecStd	108	23	18	23	17	34	28	40	35	32	24	24	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	12.8	16.3	24.5	27.3	31.6	32.2	32.3	26.6	20.9	12.9	11.2
		MCWB	8.6	9.6	10.7	15.1	16.9	20.0	18.4	19.7	17.5	16.3	10.9	9.4
	2%	DB	8.8	11.1	13.3	20.0	24.7	27.6	30.0	29.1	23.7	17.5	11.3	9.2
		MCWB	7.1	8.6	9.4	13.1	16.4	18.4	18.8	18.7	16.6	14.1	9.7	8.0
	5%	DB	7.1	9.5	11.1	17.6	21.9	24.9	27.3	27.4	21.2	15.7	10.0	7.5
		MCWB	5.6	7.6	8.8	12.1	15.1	17.2	18.3	18.0	15.6	13.2	8.5	6.5
	10%	DB	5.4	8.0	9.2	15.3	19.6	22.3	24.8	25.4	19.0	14.0	8.5	5.8
		MCWB	4.3	6.2	7.1	10.8	13.8	16.4	17.5	17.7	14.7	12.2	7.5	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.1	11.5	16.2	18.3	21.0	21.3	21.4	18.2	16.7	11.3	9.8
		MCDB	10.3	11.8	14.3	22.4	24.6	28.6	28.5	28.5	23.7	20.8	12.0	10.6
	2%	WB	7.4	9.1	10.1	14.2	17.0	19.4	20.0	20.2	17.3	14.7	10.1	7.9
		MCDB	8.6	10.8	12.7	18.9	23.4	26.2	26.1	26.1	22.4	16.8	11.0	8.9
	5%	WB	5.6	7.7	8.8	12.6	15.7	18.2	19.0	19.2	16.3	13.5	8.7	6.6
		MCDB	6.9	9.2	11.0	16.7	20.7	23.3	25.0	24.6	20.4	15.2	9.7	7.5
	10%	WB	4.4	6.3	7.3	11.1	14.5	17.0	18.0	18.2	15.1	12.5	7.5	4.8
		MCDB	5.4	7.8	8.9	14.7	18.5	21.4	23.4	23.5	17.9	13.9	8.4	5.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.4	8.6	9.3	9.3	9.3	9.7	7.8	6.5	4.5	4.0
		MCDBR	6.1	6.5	9.2	12.0	13.1	13.5	13.5	13.2	11.2	8.2	5.8	5.0
		MCWBR	4.9	4.8	6.2	6.6	6.8	6.2	5.1	5.0	5.6	5.3	4.5	4.4
	5% WB	MCDBR	5.8	5.6	8.4	11.1	11.6	12.0	11.3	11.2	10.0	7.7	5.8	5.0
		MCWBR	4.9	4.4	6.0	6.3	6.4	6.3	5.1	5.0	5.4	5.3	4.9	4.6
Clear-Sky Solar Irradiance	taub	0.314	0.337	0.376	0.407	0.407	0.403	0.423	0.416	0.380	0.360	0.336	0.309	
	taud	2.358	2.340	2.249	2.223	2.263	2.296	2.249	2.284	2.366	2.396	2.395	2.363	
	Ebn at Noon	682	772	806	826	845	851	825	809	798	736	649	621	
	Edh at Noon	61	82	109	126	127	124	128	118	97	78	60	52	
All-Sky Solar Radiation	RadAvg	0.67	1.36	2.38	3.95	4.87	5.19	4.88	4.21	3.05	1.72	0.80	0.50	
	RadStd	0.08	0.23	0.28	0.51	0.58	0.42	0.64	0.35	0.36	0.20	0.11	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.48	N/A	N/A	N/A	N/A

Nomenclature: See separate page